Ubuntu Braai & Lounge
Website Architecture & Challenges

Overview

The Ubuntu Braai & Lounge website uses a layered architecture that separates the customer interface, website hosting and future backend services. The solution is designed to be reliable, affordable, scalable and suitable for a small business using AWS Free Tier and pay-as-you-go services.

Current Architecture Components

Component	Purpose
Customer	Browses the menu, places orders and makes reservations.
Web Browser	Displays the website and sends requests.
HTML5	Provides the structure of each page.
CSS3	Creates the responsive layout and branding.
JavaScript	Handles navigation, shopping cart, validation and local storage.
Amazon S3	Hosts the static website files at low cost.

Future AWS Services

Service	Purpose
Amazon Cognito	Customer and staff login.
API Gateway	Receives secure requests from the website.
AWS Lambda	Processes orders and reservations.
Amazon DynamoDB	Stores orders and reservations using pay-as-you-go pricing.
Amazon SES	Sends confirmation emails.
Amazon SQS	Queues orders during busy periods.
Amazon CloudFront	Improves website loading speed.
AWS Budgets	Tracks monthly AWS spending.
Amazon CloudWatch	Monitors the application and logs errors.

Challenges and AWS Solutions

Challenge	AWS Services	How the Challenge is Addressed
High order volumes	SQS, Lambda	Queues requests and processes them automatically.
Menu updates	Amazon S3	Website files are updated quickly at low cost.
Reservation conflicts	Lambda, DynamoDB	Checks availability before confirming bookings.
Customer communication	Amazon SES	Sends affordable confirmation emails.
Static website limitations	API Gateway, Lambda	Adds dynamic functionality only when needed.

Security	IAM	Restricts access to AWS resources.
Performance	CloudFront	Improves loading speed by caching content.
Cost management	AWS Budgets	Alerts when spending approaches the budget.
Monitoring	CloudWatch	Tracks health and application logs.
Backup & recovery	S3 Versioning	Recovers deleted or overwritten files.

POPIA Compliance

To help protect customers' personal information, the website would use HTTPS with AWS Certificate Manager (ACM) to secure data sent over the internet, IAM to control who can access AWS resources, encryption on Amazon S3 to protect stored files, and Amazon CloudWatch to monitor activity. These affordable services would support POPIA requirements without increasing costs unnecessarily.

Conclusion

By combining secure authentication, monitoring, scalable serverless services and low-cost storage, the Ubuntu Braai & Lounge solution addresses the identified business and technical challenges while remaining affordable for a small business. The selected AWS services would provide a practical foundation that can be expanded as the restaurant grows.